GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY

B.TECH. COMPUTER SCIENCE & ENGINEERING / ALLIED BRANCHES

END SEMESTER EXAMINATION (2023-24) -- SEMESTER 1

COURSE: ENGINEERING MATHEMATICS - I (ETMA-103)

Time Allowed: 3 Hours Maximum Marks: 70 Marks Paper Code: ETMA-103

Roll Number: [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] | Total Printed Pages: 2

NOTE: Attempt all Sections. Assume suitable missing data if any. Programmable calculators are NOT permitted.

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SECTION A: Short Answer Type Questions (Attempt ALL questions) [10 x 2 = 20 Marks]

Q1. (a) State Cayley-Hamilton theorem and write its two main applications.

Q1. (b) Define rank of a matrix and give condition for consistency of AX = B.

Q1. (c) State Euler theorem on homogeneous functions of degree n.

Q1. (d) What is an orthogonal matrix? Write its determinant property.

Q1. (e) Define Jacobian of functions u(x, y) and v(x, y).

Q1. (f) Find the eigenvalues of a 2x2 matrix with trace 5 and determinant 6.

Q1. (g) State Rolle's Mean Value Theorem with geometric interpretation.

Q1. (h) Evaluate curl of gradient of a scalar field phi.

Q1. (i) Find the stationary points of f(x, y) = x^2 + y^2 - 4x + 6y + 1.

Q1. (j) State Gauss Divergence Theorem in vector calculus.

SECTION B: Analytical & Computational Problems (Attempt any THREE) [3 x 10 = 30 Marks]

Q2. (a) Using Cayley-Hamilton theorem, find A^(-1) and A^4 for a given 3x3 matrix.

Q2. (b) Verify Euler theorem for u = sin^(-1)((x + y) / (sqrt(x) + sqrt(y))).

Q2. (c) Find the maximum and minimum distances from origin to the surface z^2 = xy + 1.

Q2. (d) Verify Green's theorem in a plane for a vector field along a rectangular contour.

Q2. (e) Diagonalize the symmetric matrix A and find the modal matrix P.

SECTION C: Long Technical & Design Questions (Attempt any ONE from each) [5 x 10 = 50 Marks]

Q3. (a) Solve the system of linear equations using Gauss Elimination method with back-substitution. [10 Marks]

-- OR --

(b) Find eigenvalues and eigenvectors of a 3x3 real symmetric matrix. [10 Marks]

Q4. (a) Expand f(x, y) = e^x cos(y) in powers of x and y using Taylor series up to 3rd degree. [10 Marks]

-- OR --

(b) If u = x(1 - y^2)^0.5 + y(1 - x^2)^0.5, prove the partial derivative relationship. [10 Marks]

Q5. (a) Find maxima and minima of f(x, y) = x^3 + y^3 - 3axy using second derivative test. [10 Marks]

-- OR --

(b) Find the volume of the ellipsoid x^2/a^2 + y^2/b^2 + z^2/c^2 <= 1 using triple integrals. [10 Marks]

Q6. (a) Verify Stokes' Theorem for F = (2x - y)i - yz^2 j - y^2 z k over upper hemisphere. [10 Marks]

-- OR --

(b) Evaluate surface integral using Gauss Divergence Theorem over a closed cylinder. [10 Marks]

Q7. (a) Find the directional derivative of phi = 2xy + z^2 at (1, -1, 3) along vector 2i + 2j - k. [10 Marks]

-- OR --

(b) Prove that curl(curl F) = grad(div F) - Laplacian(F) for any vector field F. [10 Marks]

EXAMINATION EVALUATION SCHEME & RUBRIC GUIDELINES:

1. Section A: Direct definition, time complexity analysis, formula derivation, or syntax verification.
2. Section B: Step-by-step mathematical proof, trace tables, algorithmic pseudocode, and state diagrams.
3. Section C: Architectural design, full code implementation, edge case handling, and space/time tradeoffs.

***** END OF QUESTION PAPER -- DEVENGERS ACADEMIC VAULT *****